

Tracing cosmic structure post-reionization with neutral hydrogen



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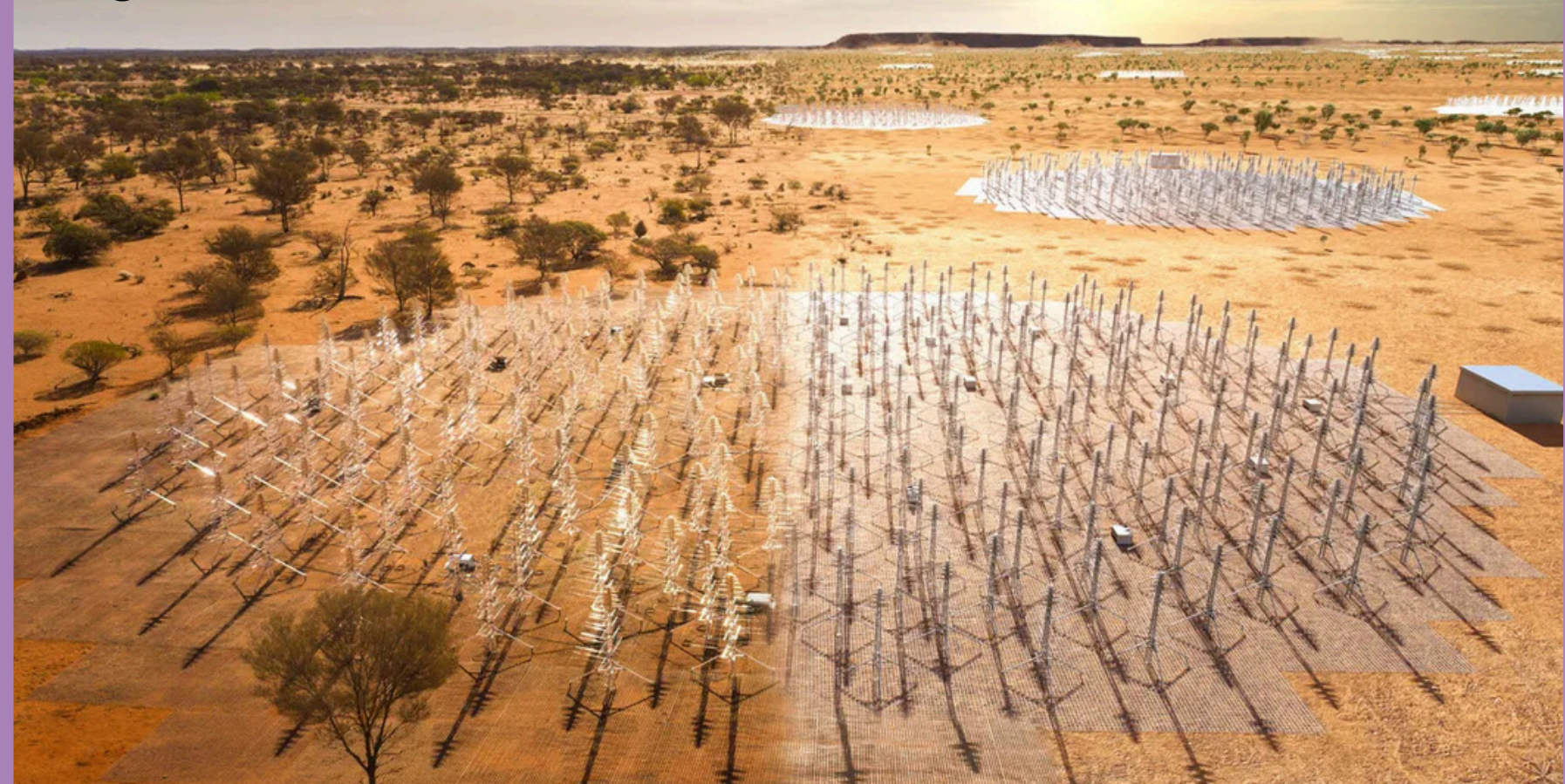
Background

- The **Epoch of Reionization** (EoR) is the era of the universe where ionizing photons produced by stars and galaxies gradually reionize the neutral **intergalactic medium** (IGM).
- In this work, we are focusing on the **transition** from the EoR to late time structure, covering redshifts $3 < z < 6$.
 - Reionization concludes**
 - IGM completely ionized
- Neutral hydrogen** (HI) remains only in small islands towards the centre of halos.
 - Medium is **sufficiently dense** to **self-shield** from ionizing photons.
- HI field goes from a tracer of the astrophysics of reionization to a **tracer of the cosmic web**.
- The upcoming Square Kilometre Array (SKA) will be able to observe in this redshift range.

Aims

- Effectively** measure the HI signal during the **post-EoR** transition to large-scale structure.
- Examine the **detectability** of this **transition** signal with **SKA-Low**.
- Evaluate the **optimal frequency binning** for balancing **thermal noise** with accessible **line-of-sight scales**.

Image credit: SKAO



Methods

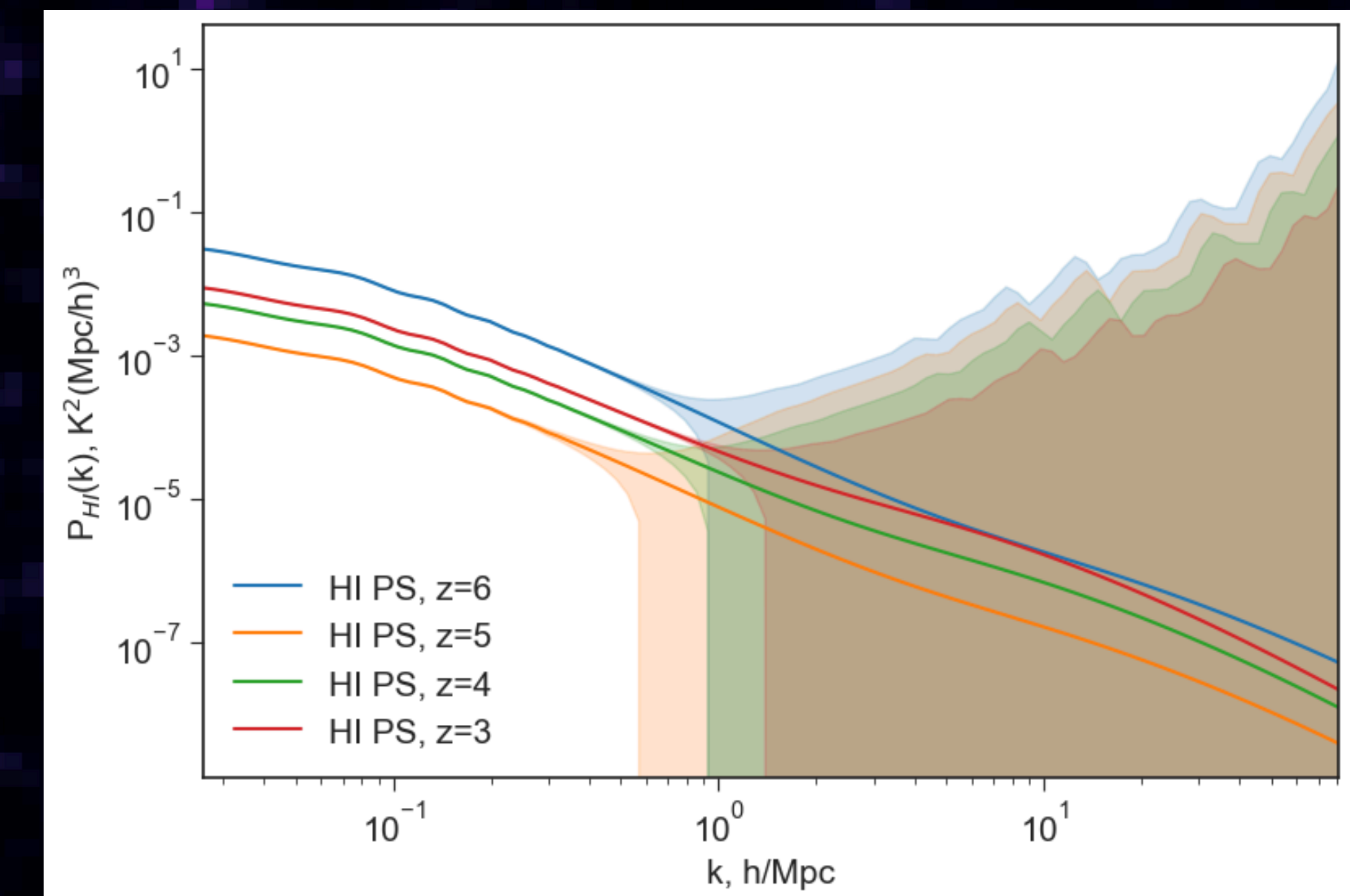
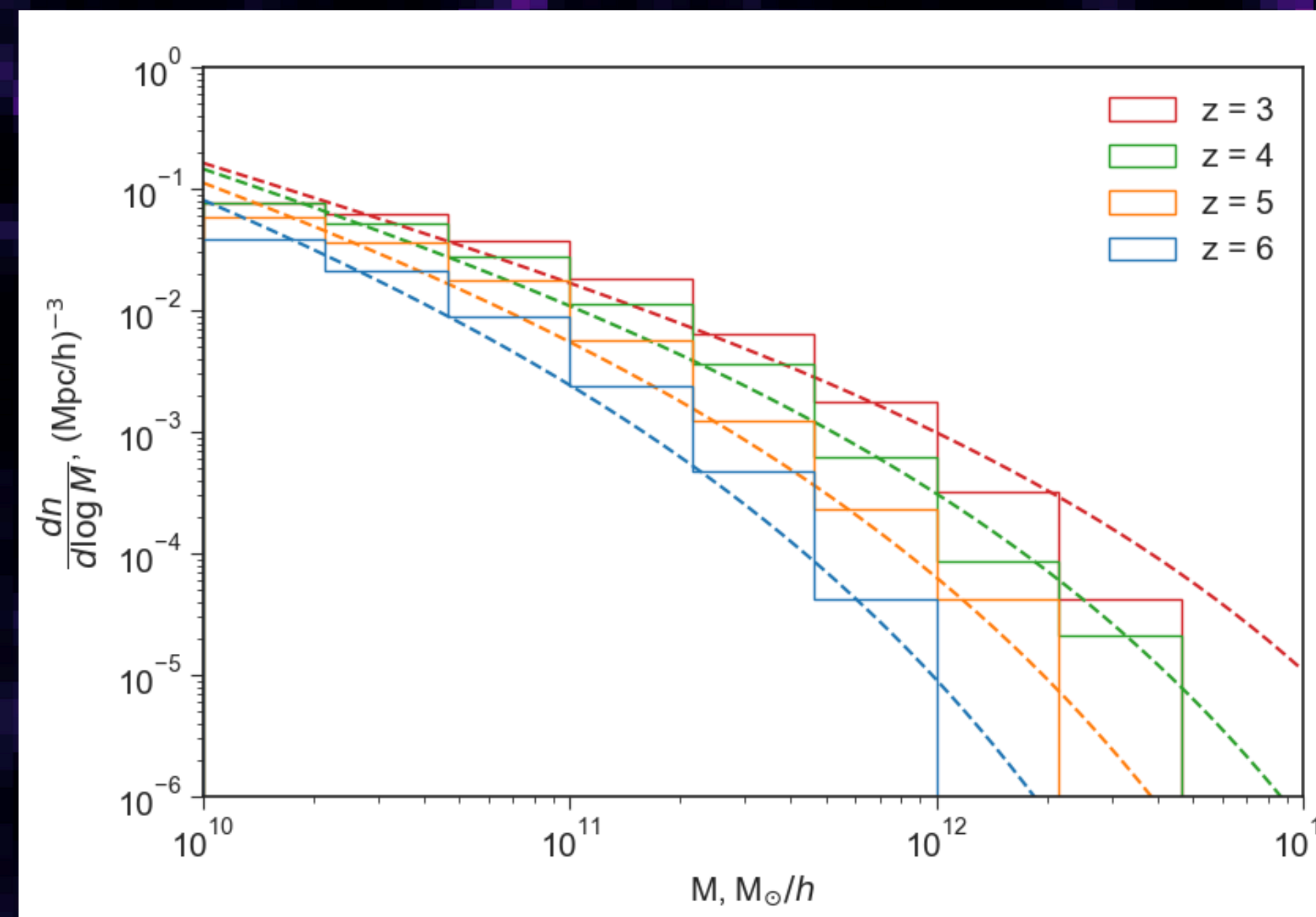
- We build off of the EoR code **21cmFAST** [1, 2].
- Apply a **watershed**-based algorithm to identify **halos**.
- Use a **HI-halo mass relation** and a **radial density profile** to distribute HI about each halo.
- Convert from HI density to **brightness temperature**, obtaining the observable field.

Observational effects are calculated based on the **SKA-Low deep survey**: **5000 hours** of observation over **100 deg²**.

Results

Our **halo mass function** (solid) agrees with the **theoretical Wilson** values (dotted) [3, 4] and the results of a **Friends-of-Friends** halo finder (not shown) [5].

Based on forecasts (using a non-linear power spectrum generated via [6]), the signal should be **detectable against thermal noise** up to scales of around **0.5 h/Mpc** for all redshifts.



Summary

- Successful **watershed halo finder** implementation, matching the **theoretical halo mass function**.
- Creation of a **lightcone** that concurrently models HI based on **EoR physics** (from 21cmFAST) and **late-time HI models**.
- Forecasts for **SKA-Low** suggest signal is detectable for cosmological (large) scales.

References

- [1] Mesinger, A., Furlanetto, S. and Cen, R. (2010). **21cmfast: A fast, seminumerical simulation of the high-redshift 21-cm signal**, Monthly Notices of the Royal Astronomical Society, 411(2), pp. 955–972.
- [2] Murray, S. et al. (2020). **21cmFAST V3: A python-integrated C code for generating 3D realizations of the cosmic 21cm signal**, Journal of Open Source Software, 5(54), p. 2582.
- [3] Murray, S.G., Power, C. and Robotham, A.S.G. (2013). **HMFCALC: An online tool for calculating dark matter halo mass functions**, Astronomy and Computing, 3–4, pp. 23–34.
- [4] Watson, W.A. et al. (2013). **The halo mass function through the Cosmic Ages**, Monthly Notices of the Royal Astronomical Society, 433(2), pp. 1230–1245.
- [5] Hand, N. et al. (2018). **Nbodykit: An open-source, massively parallel toolkit for large-scale structure**, The Astronomical Journal, 156(4), p. 160.
- [6] Lewis, A., & Challinor, A. (2011). **CAMB: Code for Anisotropies in the Microwave Background**, Astrophysics Source Code Library, record ascl:1102.026